KQC7016
LAB

- Group of two
- Submit all three lab reports by zipped them into a single file, save the file as

  KQC7016_Lab_TeamX.zip or .rar

Lab 1 – Web Scraping (5%)

- Web scraping is an automated method to extract large amounts of data from websites.
- These unstructured data are stored in a structured form.
- One way to scrape websites is writing your own code (using Python)
- **Ethical Issues**: To know whether a website allows web scraping or not, one can look at the website's "robots.txt" file.
  - appending "/robots.txt" to the URL that you want to scrape.
- When the code is run, a request is sent to the URL. As a response to the request, the server sends the HTML or XML page. The code then, parses the HTML or XML page, finds the data and extracts it.

Lab 1 – Web Scraping (5%)

- The basic steps:
  1. Find the URL that you want to scrape
  2. Inspecting the Page
  3. Find the data you want to extract
  4. Write the code
  5. Run the code and extract the data
  6. Store the data in the required format
- Some useful libraries:
  - Request: send HTTP requests using Python.
  - BeautifulSoup: is used to parsing HTML and XML documents
  - Pandas: Pandas is a library used for data manipulation and analysis. (cover in next chapter)

Lab 1 – Web Scraping (5%)

- Common challenges:

✓Dynamic Content – Some websites use JavaScript to load data (use Selenium).

✓CAPTCHA & IP Blocking – Some sites block scrapers (use proxies or delays).

✓Legal Restrictions – Always check robots.txt before scraping.

Lab 1 – Web Scraping (5%)

1. Study and run lab1.py
2. Run the code and examine the input for every ‘part’ of the code
   - You may need to use # to print only one part of output at a time.
3. Prepare a report to explain the function performed in this code, with the help of the print screen of the output.
4. Modify the code to perform web scraping on any other website.
   - You may add additional functions to the code.

- Lab Report 1 (with GitHubLink)

Lab 2 – EDA (5%)

- Study the world energy dataset (WorldEnergy.csv).
- Write a program Lab2.py in GitHub to perform EDA for the dataset and plot some meaningful graphs.
- In your report, explain what type of data analytics project (and the type of data analytics) can be done using this dataset.

(Note: Do not need to execute the project)

- You are encouraged to use generative AI to assist you on the coding only, but not the analysis.

- Lab Report 2 (with GitHubLink)

Lab 3 – ANOVA (5%)

- Study the world energy dataset (WorldEnergy.csv).
- Write a program Lab3.py in GitHub to demonstrate the application of ANOVA test.
- In your report, explain why these data are chosen and provide relevant discussion.
- You are encouraged to use generative AI to assist you on the coding only, but not the analysis.

- Lab Report 3 (with GitHubLink)